



Thoshith R R 250701797-2025-CSE ▾

T2**Started on** Sunday, 30 August 2026, 11:01 AM**State** Finished**Completed on** Sunday, 30 August 2026, 11:02 AM**Time taken** 32 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of $\text{arr}[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int compare(const void *a, const void *b) {
5      int x = *(int *)a;
6      int y = *(int *)b;
7
8      if (x > y)
9          return 1;
10     else if (x < y)
11         return -1;
12     else
13         return 0;
14 }
15
16 int main() {
17     int n;
18     scanf("%d", &n);
19
20     int arr[n];
21
22     for (int i = 0; i < n; i++) {
23         scanf("%d", &arr[i]);
24     }
25
26     qsort(arr, n, sizeof(int), compare);
27
28     long long sum = 0;
29
30     for (int i = 0; i < n; i++) {
31         sum += (long long)arr[i] * i;
32     }
33
34     printf("%lld", sum);
35
36     return 0;
37 }
```

	Input	Expected	Got	
✓	5 2 5 3 4 0	40	40	✓
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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